

Discovering Latent Rest-to-Task Connectivity Mappings via a Matrix-Logarithmic Covariance Regression

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GitHub: https://github.com/BXU69/bios731_final_project_xu

Abstract

We propose a Latent-State Matrix-Logarithmic Covariance Regression (LS-MLCR) model for mapping resting-state functional connectivity to task-state functional connectivity in neuroimaging data. The model applies a matrix-logarithm transformation to symmetric positive-definite (SPD) covariance matrices, fits an edge-wise linear regression in the resulting Euclidean space, and discovers latent subgroups of subjects with distinct rest-to-task mapping profiles via a finite Gaussian mixture estimated by the EM algorithm. Inference is carried out via nonparametric bootstrap parallelized as a SLURM job array on a high-performance computing cluster. A simulation study across 18 factorial scenarios ($n \in \{50, 100, 200\}$, $K \in \{2, 3, 4\}$, $p \in \{6, 15\}$, 500 replicates each) shows that the model recovers latent states with perfect clustering accuracy (ARI = 1.0) across all scenarios when K is known, and that BIC correctly selects the true K in over 92% of replicates for $n \geq 100$.

1 Introduction

Functional magnetic resonance imaging (fMRI) characterizes neural activity via blood-oxygen-level-dependent (BOLD) signals. A central quantity of interest is functional connectivity (FC)—the statistical dependence among brain regions—typically summarized as a symmetric positive-definite (SPD) covariance or correlation matrix. A fundamental question is how resting-state FC, measured while subjects lie passively in the scanner, predicts task-state FC, measured during cognitive performance. Resting-state scans are not only far cheaper to acquire, but are also feasible in populations for whom task-based protocols are impractical—including elderly individuals and those with cognitive impairment, who may be unable to reliably comply with task instructions or whose task performance differences would confound the fMRI signal [Cole et al., 2014, Vemuri et al., 2012].

Standard covariance regression models [Hoff & Niu, 2012] assume a single, homogeneous rest-to-task mapping across subjects. In practice, however, subjects may differ fundamentally in how strongly they recruit resting-state networks during task performance—reflecting biological subgroups, attentional strategies, or latent clinical phenotypes. We develop a model that simultaneously estimates the mapping and discovers these latent subgroups.

Specifically, we combine: (i) the Log-Euclidean framework [Arsigny et al., 2006] to lift SPD-valued covariance matrices into an unconstrained vector space; (ii) an edge-wise linear regression model; and (iii) a finite Gaussian mixture of regressions [Jacobs et al., 1991, Grün & Leisch, 2007] with K components estimated via the Expectation-Maximization (EM) algorithm [Dempster et al., 1977]. Bootstrap inference is parallelized using SLURM job arrays on the Emory RSPH cluster.

2 Methods

2.1 Data Setup and Notation

For n subjects indexed $i = 1, \dots, n$, we observe SPD covariance matrices $\mathbf{C}_i^{(\text{rest})}, \mathbf{C}_i^{(\text{task})} \in \text{Sym}^+(p)$ (the space of $p \times p$ symmetric positive-definite matrices). We apply the matrix logarithm to map each into the tangent space at the identity:

$$\mathbf{L}_i^{(\cdot)} = \log(\mathbf{C}_i^{(\cdot)}) = \mathbf{U} \text{diag}(\log \lambda_1, \dots, \log \lambda_p) \mathbf{U}^\top,$$

where $\mathbf{C} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ is the eigendecomposition. We then half-vectorize the upper triangle (including diagonal) via $\text{vech}(\cdot)$, yielding $q = p(p+1)/2$ -dimensional vectors $\mathbf{r}_i = \text{vech}(\mathbf{L}_i^{(\text{rest})})$ (predictor) and $\mathbf{s}_i = \text{vech}(\mathbf{L}_i^{(\text{task})})$ (response).

2.2 Model Specification

We assume K latent states with mixing proportions $\pi_k > 0$, $\sum_k \pi_k = 1$, and latent assignment $z_i \in \{1, \dots, K\}$ with $\Pr(z_i = k) = \pi_k$. Conditional on $z_i = k$, the task connectivity vector follows an edge-wise Gaussian regression:

$$\mathbf{s}_i \mid z_i = k \sim \mathcal{N}_q(\boldsymbol{\beta}_0^{(k)} + \text{diag}(\boldsymbol{\beta}_1^{(k)})\mathbf{r}_i, \text{diag}(\boldsymbol{\sigma}^{2(k)})), \quad (1)$$

where $\boldsymbol{\beta}_0^{(k)}, \boldsymbol{\beta}_1^{(k)} \in \mathbb{R}^q$ are state-specific intercept and slope vectors, and $\boldsymbol{\sigma}^{2(k)} \in \mathbb{R}_+^q$ is a diagonal residual variance. The diagonal covariance means each edge j is modeled by an independent univariate regression; the full-data marginal is a K -component Gaussian mixture:

$$p(\mathbf{s}_i \mid \mathbf{r}_i, \boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathcal{N}_q(\mathbf{s}_i; \boldsymbol{\beta}_0^{(k)} + \text{diag}(\boldsymbol{\beta}_1^{(k)})\mathbf{r}_i, \text{diag}(\boldsymbol{\sigma}^{2(k)})),$$

where $\boldsymbol{\theta} = \{\pi_k, \boldsymbol{\beta}_0^{(k)}, \boldsymbol{\beta}_1^{(k)}, \boldsymbol{\sigma}^{2(k)}\}_{k=1}^K$ denotes the collection of all model parameters.

2.3 EM Algorithm

The observed-data log-likelihood is $\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \log[\sum_k \pi_k \phi_q(\mathbf{s}_i; \boldsymbol{\mu}_i^{(k)}, \text{diag}(\boldsymbol{\sigma}^{2(k)}))]$, where $\boldsymbol{\mu}_i^{(k)} = \boldsymbol{\beta}_0^{(k)} + \text{diag}(\boldsymbol{\beta}_1^{(k)})\mathbf{r}_i$, and $\phi_q(\cdot; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the q -dimensional multivariate Gaussian density with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. The EM algorithm alternates between:

E-step. Compute posterior responsibilities $\gamma_{ik} = \frac{\pi_k \phi_q(\mathbf{s}_i; \boldsymbol{\mu}_i^{(k)}, \text{diag}(\boldsymbol{\sigma}^{2(k)}))}{\sum_{\ell=1}^K \pi_\ell \phi_q(\mathbf{s}_i; \boldsymbol{\mu}_i^{(\ell)}, \text{diag}(\boldsymbol{\sigma}^{2(\ell)}))}$, evaluated via the log-sum-exp trick to avoid floating-point underflow.

M-step. With $N_k = \sum_i \gamma_{ik}$, update in closed form: $\hat{\pi}_k = N_k/n$; for each edge j and state k , weighted least-squares (WLS) gives $\hat{\beta}_{1j}^{(k)}$ and $\hat{\beta}_{0j}^{(k)}$ (see [Appendix B](#)); residual variances follow from weighted squared residuals, floored at 10^{-6} .

Initialization. We run $n_{\text{init}} = 20$ independent EM runs, each initialized by applying K -means clustering to the response vectors $\{\mathbf{s}_i\}_{i=1}^n$ followed by within-cluster WLS, and return the run with the highest final log-likelihood. This multi-start strategy substantially reduces sensitivity to poor initialization.

Convergence. We declare convergence when the relative log-likelihood increment falls below $\varepsilon = 10^{-4}$, with a maximum of 500 iterations and an effective-cluster-size floor of $N_k \geq 2$ to prevent degenerate components.

Model selection. We fit $K = 1, \dots, K_{\max}$ and select via BIC: $\text{BIC}(K) = -2\ell(\hat{\theta}_K) + d_K \log n$, where $d_K = K - 1 + 2qK$ counts free parameters.

2.4 Bootstrap Inference

The EM estimator lacks a tractable analytic standard error. We use the nonparametric bootstrap: for each of B replicates, resample n subjects with replacement, warm-start EM from the full-data estimate, and record $\hat{\theta}^{*(b)}$. Standard errors and 95% percentile confidence intervals are then obtained across replicates.

Label switching. Because mixture components are identifiable only up to permutation, each bootstrap estimate must be aligned to the full-data reference before aggregating. For the b -th bootstrap replicate, we find the optimal permutation $\hat{\tau}^{(b)}$ over $\mathcal{S}_K$ (the set of all $K!$ permutations of $\{1, \dots, K\}$) by solving

$$\hat{\tau}^{(b)} = \arg \min_{\tau \in \mathcal{S}_K} \sum_{k=1}^K \left\| \hat{\beta}_1^{*(b, \tau(k))} - \hat{\beta}_1^{(k)} \right\|^2,$$

where $\hat{\beta}_1^{(k)}$ is the slope vector of the k -th component from the full-data fit, $\hat{\beta}_1^{*(b, \tau(k))}$ is the slope vector of $\tau(k)$ -th component from the b -th bootstrap replicate, and $\tau(k)$ is the permuted label assigned to reference component k . For $K \leq 4$ we enumerate all $K!$ permutations; for $K > 4$ we use greedy matching.

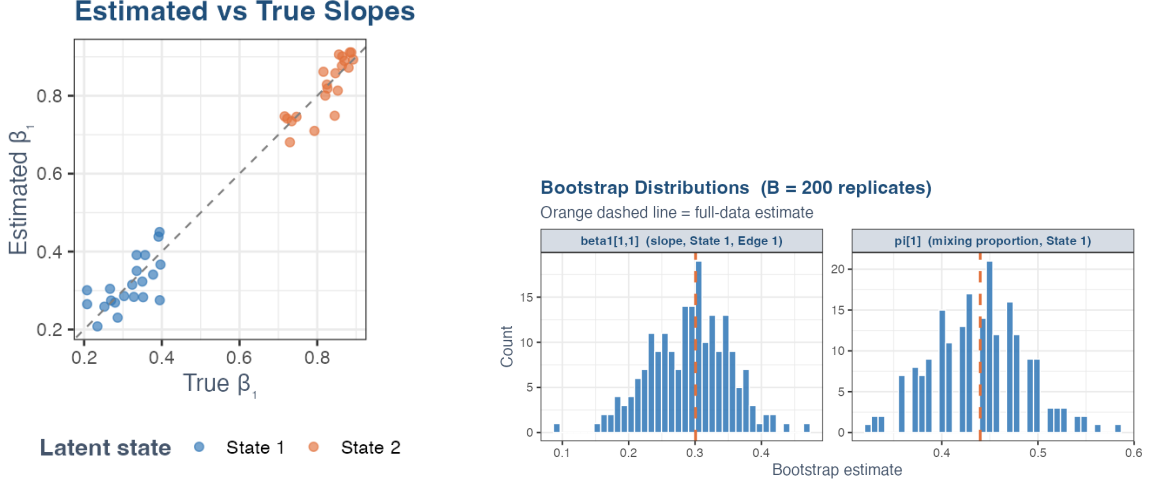
2.5 Computational Infrastructure

The bootstrap is the dominant computational bottleneck: B independent EM fits, each costing $O(nqK \cdot T)$ per bootstrap run, where T is the number of EM iterations. For $B = 1000$, $n = 100$, $q = 210$ ($p = 20$), $K = 3$, serial execution becomes prohibitive.

The simulation study parallelize across the 18 factorial scenarios ($n \in \{50, 100, 200\}$, $K \in \{2, 3, 4\}$, $p \in \{6, 15\}$) on the Emory RSPH high-performance cluster using a SLURM job array (`#SBATCH --array=1-18`), where each task runs 500 Monte Carlo replicates for one scenario, consuming at most 8 GB RAM on the `wrobel` partition with `--time=UNLIMITED`. This architecture offers four key advantages over R-internal parallelism (e.g., `parallel::mclapply`):

1. **Fault isolation:** a degenerate bootstrap sample that crashes one task does not affect others.
2. **No shared-memory overhead:** each task is an independent R process with its own memory space, avoiding copy-on-write fork overhead.
3. **Scheduler integration:** SLURM handles queuing, load balancing, and time-limit enforcement transparently.
4. **Reproducibility:** each task seeds its RNG deterministically from `SLURM_ARRAY_TASK_ID`, giving exact bit-level reproducibility.

For future real-data application, the same infrastructure can be extended to parallelize bootstrap inference: $B = 1,000$ replicates can be split into batches of $b_{\text{batch}} = 10$, submitted as a 100-task SLURM array, and aggregated post hoc via `R/combine_results.R`. This would make full-scale bootstrap inference feasible for large neuroimaging cohorts where serial execution is prohibitive.



(a) Estimated vs. true slope parameters $\hat{\beta}_1$ for all $q = 21$ edges across both states (best-of-20 EM fit; replicates, $n = 100$, $p = 6$, $K = 2$). Dashed lines = $n = 100$, $p = 6$, $K = 2$. (b) Bootstrap distributions of $\hat{\pi}_1$ and $\hat{\beta}_{1,1}^{(1)}$ ($B = 200$ replicates). Orange dashed line = full-data estimate.

Figure 1: Local demo results ($n = 100$, $p = 6$, $K = 2$): parameter recovery (left) and bootstrap inference (right).

3 Simulation Study

3.1 Design

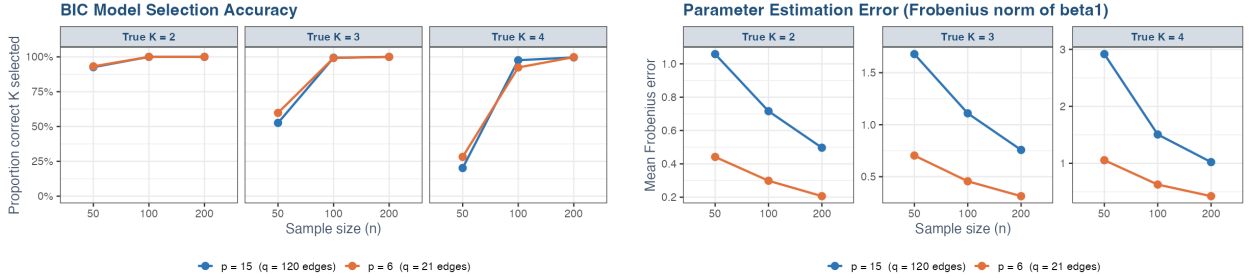
Data are generated under model (1) using the function `sim_lcr_data()` in `R/simulate_data.R`. We consider a $3 \times 3 \times 2$ factorial design: sample sizes $n \in \{50, 100, 200\}$, number of true states $K \in \{2, 3, 4\}$, and number of brain regions $p \in \{6, 15\}$ (yielding $q \in \{21, 120\}$ edges). The $p = 6$ setting matches a local sanity check; $p = 15$ ($q = 120$) represents a higher-dimensional regime to assess how algorithm performance scales with dimensionality. For each of the 18 scenarios, we draw 500 Monte Carlo replicates. Performance is evaluated on four metrics: BIC model-selection accuracy, mean Frobenius estimation error $\|\hat{\beta}_1 - \beta_1^{\text{true}}\|_F$, Adjusted Rand Index (ARI) between estimated and true state assignments, and convergence rate.

3.2 Results

Local demo. On a single dataset ($n = 100$, $p = 6$, $K = 2$) with 20 random restarts, the EM algorithm converges rapidly and monotonically in a small number of iterations regardless of initialization, with the best-of-20 run consistently attaining a higher log-likelihood than any single random start. Figure 1a shows that all $2q = 42$ estimated slope parameters $\hat{\beta}_1$ lie tightly along the identity line, confirming accurate recovery of state-specific rest-to-task mappings. Estimated mixing proportions closely match the truth, and bootstrap inference (Figure 1b) yields narrow, symmetric confidence intervals consistent with asymptotic normality of the EM estimator.

Full simulation. Summary statistics across all 18 scenarios are shown in Table 1 (Appendix A). ARI equaled 1.0 in every scenario and replicate, confirming that, given the true K , the EM algorithm perfectly recovers true state assignments across all sample sizes and dimensionality settings. BIC accuracy scaled with n : at $n = 100$ it exceeded 92% in all settings, and at $n = 200$ it reached $\geq 99\%$ across scenarios (Figure 2a). At the smallest sample size ($n = 50$), accuracy degraded markedly

for $K \geq 3$: 59.8% for $K = 3$, $p = 6$ and only 20.2% for $K = 4$, $p = 15$, reflecting BIC’s well-known tendency to under-select the number of components when the penalty dominates at small n . The Frobenius estimation error (Figure 2b) decreased monotonically with n in every scenario and was larger at higher dimensionality ($p = 15$, $q = 120$) than at lower dimensionality ($p = 6$, $q = 21$), consistent with the $O(\sqrt{qK/n})$ scaling of weighted regression estimation error. Error also increased with K (Table 1), as expected from the larger number of free parameters and harder initialization. The EM algorithm converged in all replicates across all 18 scenarios.



(a) BIC model-selection accuracy by n , p , and K . (b) Mean Frobenius estimation error $\|\hat{\beta}_1 - \beta_1^{\text{true}}\|_F$ vs. sample size, stratified by dimensionality and K .

Figure 2: Full simulation results (500 replicates per scenario): BIC model selection accuracy (left) and parameter estimation error (right).

4 Discussion

We proposed and validated the LS-MLCR model for discovering latent subgroups with distinct rest-to-task FC mapping profiles. The simulation study confirms several key properties. First, ARI was exactly 1.0 across all 18 scenarios, indicating that when the number of states K is known, the EM algorithm perfectly recovers true state assignments across the sample sizes and dimensionality settings considered here. Second, BIC selection requires at least $n = 100$ subjects for reliable K identification, and struggles at $n = 50$ when $K \geq 3$; this provides a practical power guideline for future neuroimaging studies.

A natural next step is to apply the proposed model to a real resting-state and task-state fMRI dataset, which remains the primary direction for future work. However, the proposed model may only be applicable to low- to moderate- dimensional real data. The current implementation may not scale to high-dimensional real datasets with large p : the number of edges $q = p(p + 1)/2$ grows quadratically with the number of brain regions, and both the M-step and bootstrap become computationally expensive when q is large; dimension reduction or sparse parameterization of β_1 would be needed in such settings. Beyond scalability, the diagonal residual covariance in (1) ignores within-state cross-edge correlations present in real brain networks; a low-rank or sparse off-diagonal extension could improve calibration. Second, the number of states K must be prespecified and grid-searched via BIC; a Dirichlet process mixture would allow K to grow adaptively with n . Third, graph-structural regularization (e.g., graphical lasso penalties on $\hat{\beta}_1$) could further exploit the known topology of functional brain networks.

References

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Appendix A Simulation Study Summary Table

Table 1: Simulation study summary (500 replicates per scenario). Columns: BIC accuracy, mean Frobenius estimation error in $\hat{\beta}_1$, mean ARI, and convergence rate (%).

K	p	n	BIC Acc.	$\hat{\beta}_1$ Frobenius err.	ARI	Conv. (%)
2	6	50	0.932	0.442	1.000	100
		100	1.000	0.299	1.000	100
		200	1.000	0.206	1.000	100
	15	50	0.926	1.059	1.000	100
		100	1.000	0.716	1.000	100
		200	1.000	0.497	1.000	100
	6	50	0.598	0.704	1.000	100
		100	0.992	0.456	1.000	100
		200	1.000	0.313	1.000	100
3	6	50	0.598	0.704	1.000	100
		100	0.992	0.456	1.000	100
		200	1.000	0.313	1.000	100
	15	50	0.526	1.679	1.000	100
		100	0.994	1.110	1.000	100
		200	1.000	0.757	1.000	100
	6	50	0.282	1.055	1.000	100
		100	0.924	0.627	1.000	100
		200	0.998	0.424	1.000	100
4	6	50	0.282	1.055	1.000	100
		100	0.924	0.627	1.000	100
		200	0.998	0.424	1.000	100
	15	50	0.202	2.917	1.000	100
		100	0.976	1.506	1.000	100
		200	0.996	1.022	1.000	100

Appendix B M-Step Closed-Form Updates

The complete-data log-likelihood decomposes over states k and edges j :

$$\ell_c(\boldsymbol{\theta}) = \sum_{k=1}^K \sum_{j=1}^q \sum_{i=1}^n \gamma_{ik} \left[\log \pi_k - \frac{1}{2} \log \sigma_j^{2(k)} - \frac{(s_{ij} - \beta_{0j}^{(k)} - \beta_{1j}^{(k)} r_{ij})^2}{2\sigma_j^{2(k)}} \right].$$

Treating γ_{ik} as fixed (from the E-step), maximizing over $\beta_{0j}^{(k)}$ and $\beta_{1j}^{(k)}$ for each (k, j) pair yields weighted least squares. Define $N_k = \sum_i \gamma_{ik}$, $\bar{r}_{kj} = N_k^{-1} \sum_i \gamma_{ik} r_{ij}$, $\bar{s}_{kj} = N_k^{-1} \sum_i \gamma_{ik} s_{ij}$. Then:

$$\hat{\beta}_{1j}^{(k)} = \frac{\sum_i \gamma_{ik} (r_{ij} - \bar{r}_{kj})(s_{ij} - \bar{s}_{kj})}{\sum_i \gamma_{ik} (r_{ij} - \bar{r}_{kj})^2}, \quad (2)$$

$$\hat{\beta}_{0j}^{(k)} = \bar{s}_{kj} - \hat{\beta}_{1j}^{(k)} \bar{r}_{kj}, \quad (3)$$

$$\hat{\sigma}_j^{2(k)} = \max\left(\frac{\sum_i \gamma_{ik} (s_{ij} - \hat{\beta}_{0j}^{(k)} - \hat{\beta}_{1j}^{(k)} r_{ij})^2}{N_k}, 10^{-6}\right). \quad (4)$$

These $K \times q$ subproblems are mutually independent and can be computed in a single vectorized pass. In R, exploiting `colSums` and broadcasting via `sweep` reduces the M-step to a small number of matrix operations, making it the cheapest part of each EM iteration despite the large parameter count.

Appendix C Label-Switching Resolution

For $K \leq 4$ components, we enumerate all $K!$ permutations of $\{1, \dots, K\}$ and select the permutation $\hat{\tau}$ minimizing the squared Frobenius distance between the permuted bootstrap slope matrix and the reference. For $K > 4$ (not used in the current study), we replace exhaustive search with a greedy matching: iteratively assign each reference component to the closest unmatched bootstrap component based on squared Euclidean distance in $\hat{\beta}_1$ space. Both procedures are implemented in `R/em.functions.R (align_labels)`.